Building the Future: Leveraging Artificial Intelligence (AGI) in the Construction Industry

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Introduction

Artificial intelligence (AI) has already changed many industries by automating repetitive tasks, analyzing large amounts of data, and helping professionals make faster decisions. However, most AI systems today are considered "narrow AI" because they are designed to perform specific tasks and cannot think or adapt beyond their programmed purpose. Artificial General Intelligence (AGI) represents the next stage of artificial intelligence. Unlike traditional AI, AGI would be capable of learning, reasoning, and solving problems across many different situations in a way that is much closer to human intelligence (IBM, 2023).

The construction industry is one of the largest and most important industries in the world, yet it continues to face challenges such as labor shortages, rising material costs, project delays, safety risks, and communication issues between contractors and project teams. While current AI tools are already helping improve certain aspects of construction, they are limited to individual tasks and require significant human oversight. AGI has the potential to take these capabilities much further by managing complex projects, adapting to unexpected challenges, and coordinating multiple aspects of construction at the same time (McKinsey & Company, 2024).

This case study explores how Artificial General Intelligence could transform the construction industry by improving project planning, increasing worker safety, reducing costs, and streamlining communication throughout the construction process. It also examines the ethical concerns and potential risks associated with implementing AGI, including workforce changes, cybersecurity, accountability, and the continued need for human oversight.

As shown in **Figure 1**, AGI could provide construction teams with real-time project information, interactive digital building models, and intelligent decision support directly at the job site, allowing workers and managers to collaborate more efficiently throughout every stage of a project.

Industry Analysis

The construction industry plays a vital role in building the homes, businesses, roads, bridges, schools, and infrastructure that communities rely on every day. As populations continue to grow and cities expand, the demand for new construction projects continues to increase. Despite its importance, construction remains one of the most complex industries to manage because every project involves multiple teams, strict safety regulations, changing customer requirements, and large financial investments. Even small communication errors or unexpected delays can significantly increase both project costs and completion times (McKinsey & Company, 2024).

One of the biggest challenges facing the construction industry today is project scheduling and coordination. Construction projects often involve architects, engineers, subcontractors, suppliers, inspectors, and project managers who must work together while meeting strict deadlines. Delays in material deliveries, labor shortages, severe weather, equipment failures, or design changes can quickly disrupt an entire project schedule. In addition, rising material costs and increasing labor shortages have made it even more difficult for companies to complete projects on time and within budget.

Safety also remains a major concern throughout the industry. Construction workers operate heavy machinery, work at significant heights, and regularly face hazardous conditions that increase the risk of workplace injuries. According to the Occupational Safety and Health Administration (OSHA, 2025), falls, electrocutions, struck-by incidents, and caught-in or between accidents continue to be among the leading causes of fatalities in construction. As projects become larger and more complex, maintaining safe working environments becomes increasingly challenging.

Artificial intelligence has already begun improving several aspects of construction. Companies currently use AI-powered software to analyze project schedules, monitor construction progress through drones and cameras, optimize Building Information Modeling (BIM), predict equipment maintenance needs, and improve worker safety by identifying hazards before accidents occur (Autodesk, 2024). While these technologies have produced measurable improvements, they are still designed to perform specific tasks and cannot independently reason through complex situations or coordinate every part of a project. Artificial General Intelligence has the potential to move beyond these limitations by serving as an intelligent decision-making partner capable of managing multiple aspects of construction simultaneously.

What is Artificial General Intelligence (AGI)?

Artificial General Intelligence (AGI) refers to a theoretical form of artificial intelligence that can learn, reason, adapt, and solve problems across many different situations rather than being limited to a single task. Unlike the AI systems used today, which are designed for specific purposes such as recognizing images, analyzing schedules, or detecting safety hazards, AGI would be capable of understanding new information, applying knowledge from one situation to another, and making decisions with minimal human guidance (IBM, 2023).

Today's artificial intelligence is often called narrow AI because it performs one specialized function at a time. For example, one AI program may analyze blueprints, while another predicts equipment maintenance or identifies safety risks on a construction site. Although these systems can be highly effective, they cannot combine knowledge from multiple areas or independently adjust to unexpected situations outside of their programming.

Table 1

Comparison of Artificial Intelligence and Artificial General Intelligence

Note. This table compares the primary characteristics of today's narrow AI systems with the broader capabilities expected from future Artificial

Artificial Intelligence (AI)	Artificial General Intelligence (AGI)
Designed for specific tasks	Capable of learning across many tasks
Requires separate systems for different functions	Integrates multiple functions into one intelligent system
Limited ability to adapt to unfamiliar situations	Can reason and adapt to new changes
Requires frequent human guidance	Makes informed recommendations with minimal supervision
Focuses on narrow applications	Understands broader project goals and relationships

AGI would represent a significant advancement because it could integrate all of these capabilities into a single intelligent system. Instead of relying on separate software programs, an AGI system could simultaneously coordinate project schedules, monitor worker safety, manage supply chains, analyze building designs, estimate costs, and respond to unexpected challenges in real time. This ability to reason across multiple domains would make AGI a valuable decision-support tool for complex industries such as construction.

Rather than replacing experienced engineers, architects, or construction professionals, AGI would likely serve as an intelligent partner by providing recommendations, identifying potential problems before they occur, and helping project teams make faster and more informed decisions. Human expertise would remain essential, while AGI would enhance efficiency, communication, and overall project management.

AGI Application Proposal

Artificial General Intelligence has the potential to transform nearly every stage of the construction process by serving as an intelligent assistant that continuously learns, adapts, and supports decision-making. Unlike today's AI systems, which perform isolated tasks, AGI could analyze information from multiple sources simultaneously and coordinate entire projects from planning to completion. By combining real-time data with human expertise, AGI could help construction teams make faster, more informed decisions while reducing delays, improving safety, and increasing overall efficiency (World Economic Forum, 2024).

One of AGI's most valuable applications would be project planning and scheduling. Construction projects often experience delays because of changing weather conditions, labor shortages, supply chain disruptions, or unexpected design modifications. An AGI system could continuously monitor these factors, predict potential scheduling conflicts before they occur, and automatically recommend alternative timelines or resource allocations. Rather than simply reacting to problems after they arise, project managers could use AGI to anticipate challenges and make proactive decisions that keep projects moving forward. However, even the most advanced AGI would not eliminate delays caused by government permitting, inspections, zoning requirements, legal disputes, or other regulatory processes that still require human approval.

AGI could also improve communication between everyone involved in a construction project. Architects, engineers, subcontractors, suppliers, inspectors, and project managers frequently rely on separate software platforms and communication channels, making it easy for important information to be overlooked. An AGI system could serve as a centralized coordinator by tracking project updates, identifying inconsistencies between plans, notifying the appropriate teams of changes, and ensuring that everyone is working from the most current information. This level of coordination could reduce costly misunderstandings and improve collaboration throughout the project.

Another major advantage of AGI would be improving worker safety. By combining information from drones, cameras, wearable devices, environmental sensors, and equipment monitoring systems, AGI could identify hazardous conditions before accidents occur. For example, it could detect unstable scaffolding, monitor workers for signs of heat stress or fatigue, identify unsafe equipment operation, or recognize when safety procedures are not being followed. Instead of waiting for a supervisor to discover a problem, AGI could immediately alert workers and recommend corrective actions, helping prevent injuries before they happen.

AGI could also support sustainable construction practices by optimizing material usage and reducing waste. Construction projects frequently generate excess material because of inaccurate estimates, design revisions, or inefficient ordering practices. An AGI system could continuously analyze inventory levels, recommend more efficient material usage, and adjust purchasing decisions as project conditions change. This would not only lower construction costs but also reduce environmental impacts by minimizing unnecessary waste and improving resource management.

Benefits of Artificial General Intelligence in Construction

If successfully implemented, Artificial General Intelligence could significantly improve the construction industry by increasing efficiency, reducing costs, enhancing safety, and supporting better decision-making. Because AGI would be capable of analyzing large amounts of information in real time, construction teams could respond to changing project conditions much faster than they can today. Instead of relying solely on manual planning and human judgment, project managers would have access to intelligent recommendations based on current project data, helping them make more informed decisions throughout every phase of construction (McKinsey & Company, 2024).

Another important benefit would be improved productivity. Construction delays often occur because information is not shared quickly, schedules are not updated efficiently, or unexpected issues require time-consuming adjustments. AGI could continuously monitor project progress, identify potential problems before they become major setbacks, and recommend practical solutions that keep work moving forward. This could reduce costly delays, minimize rework, and help projects stay closer to their planned schedules and budgets.

AGI could also improve the quality of construction projects. By analyzing building designs, construction methods, and historical project data, AGI could identify potential design conflicts or structural concerns before construction begins. During construction, it could monitor workmanship, compare completed work against project specifications, and alert teams to potential quality issues before they require expensive repairs. Detecting problems earlier would improve overall project quality while reducing long-term maintenance costs.

In addition to benefiting construction companies, AGI could improve the experience of clients and building owners. More accurate schedules, better cost estimates, and improved communication would provide greater transparency throughout the construction process. Clients would have access to more reliable project updates, while contractors could build stronger relationships through increased accountability and more predictable project outcomes. As AGI technology continues to advance, construction companies that successfully integrate these systems may gain a competitive advantage by delivering projects more efficiently while maintaining high standards of quality and safety.

Risks and Ethical Considerations

Although Artificial General Intelligence has the potential to improve many aspects of the construction industry, its implementation would also introduce several challenges and ethical concerns. Like any emerging technology, AGI would require careful planning, oversight, and responsible use to ensure that its benefits outweigh its risks. Construction companies would need to balance technological innovation with human judgment, worker protection, and regulatory compliance.

One significant concern is the potential impact on employment. As AGI becomes capable of performing more planning, scheduling, monitoring, and administrative tasks, some job responsibilities could become automated. While AGI is unlikely to replace skilled trades such as electricians, plumbers, welders, or carpenters in the near future, it may reduce the demand for certain office-based or repetitive roles. Companies would need to invest in employee training and reskilling programs so workers can adapt to new technologies rather than being displaced by them.

Cybersecurity is another important consideration. An AGI system responsible for managing project schedules, financial information, equipment, and building data would become a valuable target for cyberattacks. Unauthorized access or manipulation of these systems could lead to project delays, financial losses, safety risks, or the exposure of sensitive information. Construction companies would need strong cybersecurity measures, regular system updates, and continuous monitoring to protect both project data and critical infrastructure (National Institute of Standards and Technology [NIST], 2024).

Accountability also presents a challenge. If an AGI system recommends a design change, scheduling decision, or safety procedure that ultimately contributes to a project failure or workplace accident, determining responsibility may become complicated. Questions could arise regarding whether liability belongs to the software developer, the construction company, the project manager, or another party. For this reason, human oversight should remain an essential part of construction decision-making, with AGI serving as a decision-support tool rather than the final authority.

Finally, AGI must operate within existing laws, regulations, and industry standards. While it may improve planning and coordination, it cannot bypass government permitting processes, building inspections, zoning requirements, environmental regulations, or legal approvals. Human professionals will continue to play a critical role in ensuring that construction projects comply with all applicable laws and ethical responsibilities. The most effective approach will likely combine AGI's analytical capabilities with the experience, judgment, and accountability of construction professionals.

Conclusion

Artificial General Intelligence has the potential to transform the construction industry by improving project planning, communication, safety, and overall efficiency. Unlike today's narrow AI systems, AGI would be capable of analyzing complex situations, adapting to unexpected challenges, and supporting decision-making across every stage of a construction project. These capabilities could help reduce delays, optimize resource management, improve collaboration, and enhance the quality of completed projects.

At the same time, adopting AGI would require careful consideration of its ethical and practical challenges. Concerns such as workforce displacement, cybersecurity, accountability, and regulatory compliance cannot be overlooked. Although AGI could significantly improve how construction projects are managed, human expertise and oversight will remain essential for ensuring safe, ethical, and legally compliant decisions. Rather than replacing construction professionals, AGI should be viewed as a powerful tool that supports their knowledge and experience.

As AGI technology continues to evolve, the construction industry has an opportunity to become more efficient, innovative, and sustainable. Organizations that successfully combine advanced technology with skilled professionals may be better prepared to meet the growing demands of future construction projects while maintaining high standards of safety, quality, and accountability.

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